

PREBOARD 2024-25

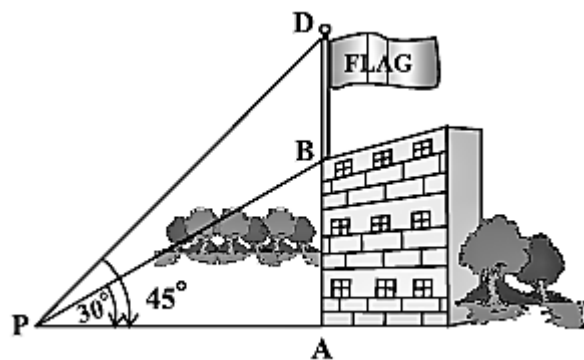
10PB24MATB03

MATHS BASIC

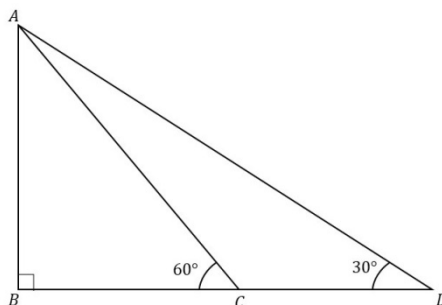
MARKING SCHEME

1	C	
2	D	
3	A	
4	D	
5	D	
6	D	
7	B	
8	B	
9	D	
10	A	
11	A	
12	B	
13	B	
14	D	
15	D	
16	A	
17	B	
18	B	
19	A	
20	D	
21	P(0,Y) section formula ratio 5;1	½,1/2,1
OR	AP = BP distance formula $(x-5)^2 + (Y-1)^2 = (X-1)^2 + (Y-5)^2$ simplification	
22	sub; values for each angle and simplifying ans;2	1 mark for accurate values & 1 mark for simplifying
23	$P(e) = 1 - P(E)$ $1/3 = 1 - P/12$ $P = 8$	½ +1+1/2
24	applying theorem and getting opp. sides equal getting all sides equal OR, correct proof with diagram	1+1 2 MARKS
25	Factorising as $(2x - 1)(x + 3)$ and $x = -5/2$ and $x = -3/2$	1 +1

	Sum as $-5/2$ and product as $-3/2$ and verifying	
26	expanding applying identities proving after simplification	1 mark 1 mark 1 mark
27	Correct proof	3 mark
28	Equating lengths of tangents and writing eqns Solving equations BL = 3cm, CM = 5cm AN = 7cm	1 mark 1 mark 1 MARK
29	$a = 2$ $a^2 = 7$ $d = 5$ $S_n = 632$ OR $a = -8d$ $a_{29} = 20d$; $a_{19} = 10d$ $a_{20} = 2(a_{19}(d))$	1+1+1 1+1+1
30	Length of the arc = $22/7$ cm Area of major sector = $726/7 \text{ cm}^2$	$\frac{1}{2} + 1$ $\frac{1}{2} + 1$
31	$3/13$, $1/52$. $1/4$	1 +1+1
32	Proof with reasoning, diagram	5 marks
33	Speed = dist/time speed be x km/h Eqn as $1500/x - 1500/x + 100 = 30/60$ Simplifying $x^2 + 100x - 300000 = 0$ $x = 500\text{km/hr}$ after discarding negative values OR fraction as $x/2x+1$ Reciprocal $2x+1/x$ Equation($x/2x+1$) + $(2x+1/x) = 58/21$ $11x^2 - 26x - 21 = 0$ Roots as 3, $-7/11$ $x = 3$ Denominator = 7 $3/7$ as the answer	1 +1+1+1+1 1+1+1+1+1
34	(a) Correct diagram	1 mark 11/2 mark 21/2 marks 1 mark 2 +2



Tan 30 and calculating distance from P to base of the building as $10\sqrt{3} = 17.32\text{m}$
 Taking tan45 and finding the length of flag staff as 7.32m
 OR



(b) correct diagram
 Taking respective values of tan 30 and tan 60 and finding the height of the tower as $20\sqrt{3}\text{ m}$

35	$H = 6\text{cm}$ Vol of solid = vol of cone + vol of hemisphere Applying correct formula and sub; Volume = 166.83 cm^3	1 +1+2+1
36	i) Here, Constant Expense = Rs. x Proportional Expense = Rs. y So, the pair of linear equations in two variables is $x + 7y = 650$(1) $x + 11y = 970$(2) ii) Subtracting Equation (1) from Equation (2), we get $11y - 7y = 970 - 650$ $4y = 320$ $y = 80$ Now putting the value of y in Equation (1), $x + 7 \times 80 = 650$ $x + 560 = 650$ $x = 650 - 560$ $= 90$ So, Fixed Expense = Rs. 90 Proportional per person expense = 80	1+2+1

	iii) For 15 guests, MrAnil has to pay $= 90 + 15 \times 80$ $= 90 + 1200$ $= \text{Rs. } 1290$		
37	(i) $\sqrt{53} \text{ units}$ (ii) $\sqrt{104} \text{ units}$ (iii)(a) $(-1, 16/5)$ OR (iii) (b) 1:1	1+1+2	
38 (i)	Class interval	Cumulative Frequency	Frequency
	15-20	2	2-0 = 2
	20-25	6	6-2 = 4
	25-30	24	24-6 = 18
	30-35	45	45-24 = 21
	35-40	78	78-45 = 33
	40-45	89	89-78 = 11
	45-50	92	92-89 = 3
	50-55	98	98-92 = 6
	55-60	100	100-98 = 2
(ii) (a)	Median = $35 + [(50-45)/33]5$ $= 35 + 25/33$ $= 35 + 0.76 = 35.76 \text{ years}$ Thus, the median age 35.76 years. Upper limit of modal class 40		
(b)	$\text{Mode} = l + \left(\frac{f_1 - f_0}{2f_1 - f_0 - f_2} \right) \times h$ $= 35 + [(33-21)/(66-21-11)]5$ $= 35 + 12 \times 5/33$ $= 35 + 1.82 = 36.82 \text{ years}$ Empirical relation, $3\text{Median} = \text{Mode} + 2\text{Mean}$ $3 \times 35.76 = 36.82 + 2M$ $107.28 = 36.82 + 2M$ $2M = 107.28 - 36.82 = 70.46$ Mean = 35.22 years		

--	--	--	--